$A$ preserves eigenspaces if $v \in V_\lambda \Rightarrow Av \in V_\lambda$

$\Leftarrow$ $Hv = \lambda v$

$\Leftarrow$ $HAv = \lambda Av = A\lambda v = AHv$

$P_\lambda = \sum_b |\lambda b\rangle\langle\lambda b|$ proj. onto $V_\lambda$

$P_\lambda |\lambda' a\rangle = \sum_b |\lambda b\rangle \langle\lambda b|\lambda' a\rangle$

$\langle\lambda b|\lambda' a\rangle = \delta_{\lambda\lambda'}\delta_{ab}$

$$P_\lambda A P_\lambda |\lambda' a\rangle = \delta_{\lambda\lambda'} P_\lambda A |\lambda' a\rangle$$

$$= \delta_{\lambda\lambda'} P_\lambda \left( \sum_{\lambda''} \sum_b A^{\lambda' a}_{\lambda'' b} |\lambda'' b\rangle \right)$$

$$= \delta_{\lambda\lambda'} \sum_{\lambda''} \sum_b A^{\lambda' a}_{\lambda'' b} \delta_{\lambda''\lambda} |\lambda'' b\rangle$$

$$= \delta_{\lambda\lambda'} \sum_b A^{\lambda' a}_{\lambda b} |\lambda b\rangle$$

① $\langle A\rangle |\lambda' a\rangle = \sum_b A^{\lambda' a}_{\lambda' b} |\lambda' b\rangle$

② $\sum_\lambda P_\lambda A P_\lambda |\lambda' a\rangle = \sum_\lambda \delta_{\lambda\lambda'} \sum_b A^{\lambda' a}_{\lambda b} |\lambda b\rangle$

$$= \sum_b A^{\lambda' a}_{\lambda' b} |\lambda' b\rangle$$

$$\boxed{\langle A\rangle = \sum_\lambda P_\lambda A P_\lambda}$$

$\rightarrow$ pointwise limit

$\longrightarrow$ $\forall \lambda', b$ $\quad \langle A\rangle |\lambda' b\rangle = \sum_\lambda P_\lambda A P_\lambda |\lambda' b\rangle$

$\langle A\rangle|_{V_\lambda} = P_\lambda A|_{V_\lambda}$ alternatively

1st act with $A$ on $V_\lambda$

and get rid of things not in $V_\lambda$

$P_\lambda |\lambda' a\rangle = \delta_{\lambda\lambda'} |\lambda' a\rangle \Rightarrow P_\lambda|_{V_{\lambda'}} = \begin{cases} \mathbb{1}|_{V_{\lambda'}} & \text{if } \lambda = \lambda' \\ 0 & \text{if } \lambda \neq \lambda' \end{cases}$

$= \delta_{\lambda\lambda'} \mathbb{1}_{V_{\lambda'}}$

$$\sum_\lambda P_\lambda A P_\lambda|_{V_{\lambda'}} = \sum_\lambda \delta_{\lambda\lambda'} P_\lambda A \mathbb{1}|_{V_{\lambda'}} = P_{\lambda'} A|_{V_{\lambda'}}$$

a projection is a map $P: W \to W$ linear s.t $P^2 = P$

$\langle A \rangle$ = proj of $A$ on something $\qquad W = \{$ linear maps $V \to V \}$

$A \in W$

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$\dim V = 3$ $\qquad$ eigen vectors $\quad \overbrace{|-1\ 1\rangle \quad |-1\ 2\rangle}^{\lambda=-1} \quad |5\ 1\rangle$

$\to \begin{pmatrix}1\\0\\0\end{pmatrix} \quad \to \begin{pmatrix}0\\1\\0\end{pmatrix} \quad \to \begin{pmatrix}0\\0\\1\end{pmatrix}$

$H|-1\ 1\rangle = -|-1\ 1\rangle$

$H|-1\ 2\rangle = -|-1\ 2\rangle$

$H|5,1\rangle = 5|5,1\rangle$

as a matrix $\quad H = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \boxed{5} \end{pmatrix}$

as a matrix $\quad A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & \boxed{i} \end{pmatrix}$

$$\langle A \rangle = \begin{pmatrix} a & b & 0 \\ d & e & 0 \\ 0 & 0 & \boxed{i} \end{pmatrix}$$

$$P_{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$P_5 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\langle A \rangle = P_{-1} A P_{-1} + P_5 A P_5$$

$$P_5 A P_5 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ g & h & i \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \boxed{i} \end{pmatrix}$$

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$$P_1 A P_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} a & b & 0 \\ d & e & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

does $[\langle A \rangle, H] = 0$ ?

$$\langle A \rangle = \left(\begin{array}{c|c} S & 0 \\ \hline 0 & T \end{array}\right) \qquad H = \left(\begin{array}{c|c} -\mathbb{1} & 0 \\ \hline 0 & s\mathbb{1} \end{array}\right)$$

$$\langle A \rangle H = \left(\begin{array}{c|c} -S & 0 \\ \hline 0 & ST \end{array}\right)$$

$$H \langle A \rangle =$$